

Environmental Health Monitoring App

Project Proposal

Developer: Laxmi Adhikari

Target Audience: Residents and local health departments.

Technology Stack: Python, Pandas, Dash, Plotly, Open-Meteo API, geopy

Background

Air quality alone does not fully represent environmental health risks. Weather conditions such as temperature, wind, humidity and precipitation significantly influence how pollutants behave and affect human health. This project expands a basic air quality monitoring system into an environmental intelligence dashboard for users. It uses the Open-Meteo API to retrieve accurate air pollution data and weather data to provide a complete understanding of outdoor environmental conditions and their health impacts.

Business Problem

Air quality and weather conditions directly affect human health, productivity, and quality of life. Despite the public availability of environmental data from various sources, that data is fragmented, raw, and inaccessible to non-technical users. People living in areas prone to heatwaves, wildfire smoke, or high particulate pollution have no easy way to understand what the numbers mean for them personally, let alone act on them in real time.

Objectives

- Develop an environmental health dashboard for users
- Integrate real-time and historical air quality data using the Open-Meteo Air quality API
- Integrate weather data (temperature, wind, humidity, precipitation) using Open-Meteo weather API
- Analyze relationship between weather conditions and pollution levels
- Provide health-focused environmental risk indicators for users

Technical Approach

1. Environmental Health Dashboard

- **Current Environmental Conditions Display:** Real-time air quality and weather information including AQI, PM2.5, PM10, temperature, humidity, precipitation, and wind speed
- **Multi-Day Forecast Integration:** Environmental outlook combining weather forecasts and predicted air quality trends.
- **AQI and Weather Code Integration:** Human-readable environmental conditions such as “Good Air Quality”, “Moderate Pollution”, “Clear Sky”, “Rainy Condition”
- **Interactive Environmental Timeline:** Visualizations showing pollution trends, temperature changes, humidity levels and weather patterns over time

2. Environment Risk & Recommendation Engine

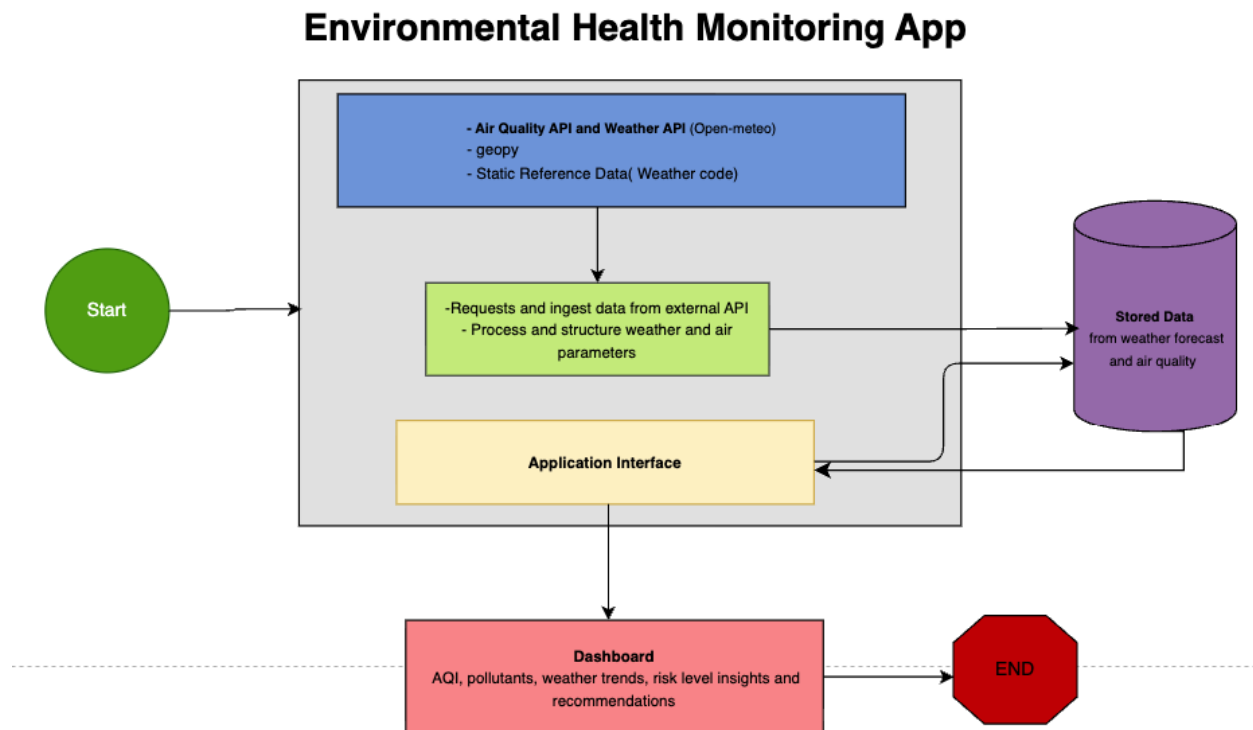
- **Health-Focused Recommendations:** Dynamic recommendations based on combined weather and air quality conditions including:
 - AQI severity levels
 - PM2.5 and PM10 concentrations
 - Temperature and humidity conditions
 - Wind speed and precipitation patterns
- **Environment Risk Categories:**

- Safe Outdoor Conditions
- Sensitive Group Warnings
- Unhealthy Air Quality Alerts
- **Recommendation Categories:**
 - Exercise Safety guidance
 - Asthma & Respiratory Precautions
 - Heat and Air pollution Awareness

3. Interactive User Interface

- **Interactive Charts:** Plotly visualizations for air quality and weather data
- **Environmental Filtering:** Dashboard updates dynamically based on selected date, pollutant type, AQI category, and weather condition using slicers
- **Health Risk Indicators:** Color-coded AQI and environmental severity indicators for easy interpretation

Technical Architecture Diagram



Backend Components

- **Data Retrieval:** Open-Meteo API integration for Air quality and weather API
- **Data Processing:** Pandas for data manipulation and transformation and merging air quality and weather datasets
- **Recommendation Engine:** Rule-based logic for generating health-focused environmental recommendations and risk indicators
- **Data Storage:** Local CSV storage for weather data and air quality data.

Frontend Components

- **Web Framework:** Dash for web application development

- **Visualization:** Plotly for interactive charts and graphs
- **UI Components:** Dash Bootstrap Components for responsive design
- **Styling:** Custom CSS for KY-themed branding

Data Flow

1. **API Integration:** Fetch air quality data and weather data from Open-Meteo API every hour
2. **Data Processing:** Transform raw API data into user-friendly formats
3. **Recommendation Logic:** Apply air quality and weather-based rules to generate activity suggestions
4. **User Interface:** Display air quality and weather data and recommendations through interactive dashboard
5. **User Interaction:** Allow users to filter environmental data pollutant, AQI level and time range

Data Sources and APIs

Primary Data Source: Open-Meteo API

- **Endpoint:**
 - <https://api.open-meteo.com/v1/forecast>
 - <https://api.open-meteo.com/v1/air-quality>
- **Location:**
 - Geopy library for dynamic search for location
- **Air Quality Parameters:**
 - PM2.5
 - PM10
 - AQI
 - Ozone(O₃)
 - Nitrogen Dioxide (NO₂)
 - Carbon Monoxide (CO)
 - Sulphur Dioxide (SO₂)
- **Weather Parameters:**
 - Current: temperature_2m, relative_humidity_2m, apparent_temperature, precipitation, weather_code, cloud_cover, wind_speed_10m
 - Daily: weather_code, temperature_2m_max/min, sunrise/sunset, precipitation_sum, wind_speed_10m_max
- **Units:**
 - Temperature: Fahrenheit
 - Wind Speed: mph
 - Precipitation: inches
 - AQI: US AQI Standard
- **Time zone:**
 - Timezone is converted to the local timezone of the selected city.

AQI and Environmental Interpretation Logic

AQI Severity Categories

- 0-50: Good air quality
- 51-100: Moderate air quality
- 101-150: Unhealthy for sensitive groups
- 151-200: Unhealthy
- 200+: Hazardous

Air quality and Weather Code Reference

- CSV-based lookup table mapping numeric codes to descriptions

- Supports multiple codes per description (comma-separated expansion)
- Enables human-readable air quality condition display

Temperature

- High temperatures may contribute to increased ozone formation
- Extreme heat conditions trigger outdoor activity cautions

Wind Speed

- Wind conditions may trap pollutants near ground level
- High wind conditions may improve pollutant dispersion

Humidity

- High humidity can affect particulate matter concentration and breathing comfort

Precipitation

- Rainfall may reduce airborne particulate pollution levels

Environmental Recommendation Logic

Recommendation Categories

- Safe Outdoor Activity Conditions
- Moderate Environmental Risk
- Sensitive Group Warnings
- Poor Air Quality Advisories

Example Recommendation Rules

- High AQI + low wind speed = pollution accumulation warning
- High temperature + elevated ozone = outdoor heat and pollution caution
- Rainfall + decreasing PM2.5 = improved air quality conditions
- AQI above unhealthy threshold = recommend limiting prolonged outdoor exposure

Responsive Design Considerations

- **Desktop:** Multi-column layout with detailed charts

Implementation Plan

Phase 1: Foundation & API Integration (Week 1)

- Set up Dash application structure
- Implement Open-Meteo API integration
- Create basic air quality data and weather data display
- Design data processing pipeline

Phase 2: Core Features (Week 2)

- Develop activity recommendation engine
- Create interactive air quality and weather visualizations
- Implement responsive UI components
- Add AQI category interpretation logic

Phase 3: Enhancement & Refinement (Week 3)

- Add weather vs pollution analysis visualization
- Implement environmental insight generation
- Optimize data processing and performance
- Add historical trend analysis
- Begin user acceptance testing

Phase 4: Testing & Optimization (Week 4)

- Comprehensive testing across devices

- Performance optimization and bug fixes
- Gather and integrate user feedback
- Finalize documentation

Phase 5: Deployment & Launch (Week 5)

- Deploy to production environment
- Monitor API reliability and performance
- Create deployment documentation
- Prepare for ongoing maintenance

Success Metrics

Technical Metrics

- **API Reliability:** >99% successful API calls
- **Page Load Time:** <3 seconds initial load
- **Data Accuracy:** Environmental data updated hourly

User Experience Metrics

- **User Engagement:** Average session duration >5 minutes
- **Recommendation Acceptance:** >70% of users interact with activity suggestions
- **Return Usage:** >50% user retention rate
- **User Satisfaction:** >4.5/5 star rating

Business Impact Metrics

- **Community Reach:** Easy to track via session analytics
- **Public Health:** Reduction in preventable outdoor exposure incidents among sensitive group (asthma, elderly)
- **Environmental Awareness:** Increase awareness of how weather pattern affects local air quality

Risk Assessment and Mitigation

Technical Risks

- **API Rate Limiting:** Implement caching and error handling
- **Data Accuracy:** Cross-reference with multiple weather and air quality sources if needed
- **Performance Issues:** Optimize data processing and implement lazy loading

Project Risks

- **Scope Creep:** Maintain clear feature prioritization
- **Timeline Delays:** Use agile development with weekly milestones
- **Resource Constraints:** Leverage open-source libraries and community resources

Budget and Resources

Development Resources

- **Libraries:** Dash, Plotly, Pandas, Requests (all free/open-source)
- **Hosting:** Heroku free tier or GitHub Pages for static deployment
- **Data:** Open-Meteo API (free, no API key required)
- **Design:** Bootstrap components and custom CSS

Time Investment

- **Development:** 5 weeks intensive development
- **Testing:** Integrated throughout each week
- **Documentation:** Ongoing throughout development
- **Maintenance:** 2-4 hours/month for updates and bug fixes

Conclusion

The Environmental Health Monitor closes a real gap in how people access and act on environmental data. Weather and air quality information has long existed in the public domain but scattered across separate APIs, expressed in technical units, and stripped of any health context. The project pulls those two streams together into a single pipeline that geocodes any location, fetches current and forecast data from Open-Meteo. By evaluating temperature, humidity, AQI, and primary pollutant concentration together, it identifies compounding conditions that neither data stream would flag alone a moderate AQI reading paired with high heat and humidity can push real-world exposure into a danger tier that a standalone weather app or AQI widget would miss entirely. Each risk classification produces specific, actionable guidance: time-of-day recommendations, mask requirements, hydration targets, and indoor air management steps calibrated to the severity of conditions on the ground.

Limitation and Future enhancements:

The current database design uses two main operational tables: `weather_observations` and `air_quality_observations`, to store weather and air quality snapshots by location and timestamp, and it is denormalized and can lead to redundancy.

In the future, a centralized locations table with relationships to both tables can be developed which will help reduce duplication and improve data integrity.

Next Steps

1. **Stakeholder Review:** Present proposal to project sponsors and stakeholders
2. **Technical Feasibility Assessment:** Verify API capabilities and library compatibility
3. **Prototype Development:** Create minimum viable product (MVP) with core features
4. **User Research:** Gather feedback on activity preferences and UI requirements
5. **Project Kickoff:** Begin Phase 1 development with clear milestones and deliverables